38 · Sperm, eggs and ovulation: the fine print

The words of this track, as it says them. Bold brown is the examiner.

Sperm, eggs and ovulation. The fine print.

The written tracks give you gamete production and the two cycles. Her quizzes ask the same ground in matching tables, cell by cell, with the chromosome counts. This track is those tables.

The hook: head, midpiece, tail. The acrosome breaks in, the mitochondria pay, and the tail swims.

The sperm cell, her six-blank table. The head holds the twenty-three chromosomes, as chromatin, and it's capped by the acrosome, which is full of enzymes to penetrate the egg. Her case: a man whose sperm are healthy, swim straight, and are plentiful, but can't penetrate an egg. The acrosome is short of the enzymes it needs. The midpiece is packed with mitochondria, the engine: they turn the fructose from the seminal vesicles into A T P, the energy for swimming, so faulty mitochondria mean poor motility. And the tail propels the sperm. Short, double or crooked tails can't. Her true or false: sperm carry a lot of food in their heads, for the long journey. False. Their fuel is the fructose in the semen.

The cells, with their chromosome counts, her matching table. The spermatogonium is the diploid stem cell, and it divides by mitosis into a diploid primary spermatocyte. That's why males make gametes for life. Meiosis one gives haploid secondary spermatocytes, and meiosis two gives haploid spermatids. Spermiogenesis turns the spermatids into spermatozoa. Her multiple choice version: spermatogonium, diploid spermatocyte, haploid spermatocyte, spermatid, spermatozoon. And inhibin, from the Sertoli cells, slows sperm production when the count is too high.

Oogenesis, her diagram. Oogonia are diploid cells that divide by mitosis into diploid primary oocytes, and that only happens in fetal life. Her true or false: oogonia are present in the ovary from menarche to menopause. False. The primary oocytes form before birth, and wait in prophase one. Meiosis one gives a haploid secondary oocyte and the first polar body. The secondary oocyte is released at ovulation, waiting in metaphase two. Meiosis two is completed at fertilisation, in telophase two, giving the ovum and a second polar body. The polar bodies are haploid cells that degenerate.

Her six differences. Duration: spermatogenesis is continuous from puberty to death, and oogenesis is monthly, from menarche to menopause. Cytokinesis: the sperm cells divide with equal cytoplasm, and the oocyte divides unequally, into one large oocyte and small polar bodies. The result: four motile sperm, against one non-motile ovum. And her false statement: both involve meiosis followed by mitosis. False. It's mitosis first, then meiosis.

Ovulation, her drop-down. A secondary oocyte is released from the ovary. Ovulation is triggered by a peak in LH. The ovarian wall ruptures when the fluid-filled Graafian follicle,

which she also calls the tertiary follicle, bursts. And the force of the fluid pushes the oocyte out, toward the fimbriae of the fallopian tube. The fimbriae sweep over the ovary to catch it, and the cilia in the tube move it on toward the uterus.

Her hormone tables. FSH, from the pituitary, is highest during the proliferative phase, and makes the follicles grow. Oestrogen, released by the developing follicle, peaks just before ovulation, and stimulates the secretion of LH. LH, from the pituitary, peaks highest just before ovulation. Progesterone, released by the corpus luteum after ovulation, dominates the secretory phase, and develops the glands of the endometrium. Oestrogen and progesterone both peak in the luteal phase. And one of her keys to learn as written: LH stimulates the production of progesterone by the follicle's thecal cells.

The traps. Oogonia only exist before birth. It's mitosis, then meiosis. The egg is released as a secondary oocyte, in metaphase two. And the acrosome gets the sperm in; the mitochondria make it swim.

Once more. Sperm, eggs and ovulation.

The sperm head holds twenty-three chromosomes as chromatin, capped by the acrosome, whose enzymes penetrate the egg. The midpiece mitochondria turn fructose into A T P for swimming, and the tail propels it. Sperm don't carry food in their heads. A diploid spermatogonium divides by mitosis into a diploid primary spermatocyte, then meiosis gives haploid secondary spermatocytes and spermatids, and spermiogenesis makes spermatozoa. Inhibin slows production when the count is high. Oogonia divide by mitosis only before birth. Meiosis one gives a haploid secondary oocyte and a polar body, and meiosis two completes at fertilisation. Polar bodies degenerate.

Spermatogenesis is continuous from puberty to death, with equal cytoplasm, and gives four motile sperm. Oogenesis is monthly, menarche to menopause, unequal, and gives one non-motile ovum. It's mitosis first, then meiosis. At ovulation, the LH peak bursts the Graafian, or tertiary, follicle, and the secondary oocyte is pushed to the fimbriae. FSH is highest in the proliferative phase. Oestrogen peaks just before ovulation and stimulates LH. Progesterone dominates the secretory phase. Her key: LH stimulates progesterone from the thecal cells.